# BASIC CONCEPTS IN ECONOMICS

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# Outline

- Economics and Economic decision-making
- Scarcity, Choice and Opportunity Cost
- The Production Possibilities Curve
- Microeconomics and the Economic Problem
- Economic Units
- Economic Systems

# Economics and Economic Decision-Making

- This unit provides an introduction to
  - the study of economics and how economic principles and models aid economic agents (including maritime economist and marine engineers) in their decision-making.
- Economics is broadly divided into two parts: microeconomics and macroeconomics
  - Microeconomics is concerned with individual parts/units of the economy
  - Macroeconomics deals with the economy as a whole

# Economics and Economic Decision-Making

- The study of economics as a subject can be traced back to the publication of a book entitled:
  - *"An inquiry into the Nature and Causes of the Wealth of Nations"* by Adam Smith in the year 1776.

- This explains why Adam Smith is often referred to as the *"father of economics"*.

- Various economists have given their own definition of economics according to their thoughts and perceptions.

# Economics and Economic Decision-Making

- Traditional definitions of economics often focused on ***wealth creation***, in order to improve the standard of living of mankind through increased consumption.

- A.C. Pigou, for example, defined it as
  - "the means of studying how total production could be increased so that the standard of living of people might be improved".

# Economics and Economic Decision-Making

- Modern definitions of economics emphasize the fundamental problem of ***scarcity*** and ***choice***.
- Key among these is the one by Lionel Robbins.
- He defined economics as
  - "the science that studies human behaviour as a relationship between ***ends*** and ***scarce means*** which have *alternative uses*".
- This definition emphasizes economics as studying the behaviour of mankind (i.e. it's a *social science*).

# Economics and Economic Decision-Making

- The definition is concerned with ***ends*** which include all the **needs**, **wants**, **targets** or **aspirations** of mankind.
- It is also concerned with **our means** (i.e. scarcity and resources).
  - The resources available are not sufficient to meet all our needs. Resources can be human (labour), natural (including land and raw materials) and manufactured inputs (capital).
- The definition recognises the fact that resources have alternative uses for people to choose from (i.e. *choice*)

# Economics and Economic Decision-Making

- Following these definitions, economics can be said to be about ***making choices under the conditions of scarcity***.
- We have to make choices because
  - the resources needed to produce goods and services are limited in supply, compared to the competing uses for them
- What is worrying in economic decision making is that *choice involves cost*
  - In choosing anything, you must be prepared to sacrifice something else

# Economics and Economic Decision-Making

- Economics can therefore be defined as the study that ascertains
  - the *most efficient allocation of scarce resources* among ***competing uses*** to ensure ***increasing production*** such that ***living standards of mankind improves*** through ***increasing consumption*.**

- The study of economics concerns maximizing society's satisfaction through the production and consumption of goods and services

# Economics and Economic Decision-Making

- Therefore, questions such as
  - how many goods and services individuals, firms and the whole economy produces?
  - how many people are employed in producing these goods and services?
  - how much do people buy and consume of these goods and services?
  - how people's consumption is affected by prices etc. are of great concern to economists?

# Scarcity, Choice and Opportunity Cost

- Everything that is considered valuable is *scarce* – *human skills*, *goods*, *time*, and *money* – whereas human wants are *unlimited.*
  - Human behaviour is such that no one is ever satisfied with what he/she has
  - Everyone at any point in time will like to have more of one thing or the other
- **Scarcity** means that human wants are ***infinite*** but the resources available to satisfy those wants are ***limited***.

# Scarcity, Choice and Opportunity Cost

- People both *rich* and *poor* want more than they can have
- However, people (individuals, firms, and the society at large) do not necessarily face the problem of scarcity to the same degree
  - A *poor man* who cannot afford basic needs like food, clothing, shelter etc. will see scarcity as more of a problem than
  - A *rich man* who cannot afford to build a second house or purchase a second Toyota Land Cruiser

# Scarcity, Choice and Opportunity Cost

- Because resources are scarce **choice** has to be made between their alternative uses.
- In any society, decisions have to be made regarding
  - *What* goods and services are going to be produced, and in what quantities?
  - *How* are the goods and services going to be produced – what production resources/techniques to use, and in what quantities?
  - *For whom* are the goods and services to be produced – how will the output be distributed?

# Scarcity, Choice and Opportunity Cost

- The choice must be rational.
- ***Rational choice*** or ***decision*** means weighing up the *costs* and *benefits* of any activity, whether it be
  - firms choosing what and how much to produce,
  - workers choosing whether to work extra hours or not, or
  - consumers choosing what to buy
- **Note:** It is rational to do any activity if the benefits exceed the cost

# Scarcity, Choice and Opportunity Cost

- Making these choices or decisions involves **sacrifice.**

- The sacrifice of alternatives (in the production or consumption of a good or service) is what economists call ***opportunity cost***.

- The ***opportunity cost*** of using any good or service is the value of the next best alternative forgone.

# Scarcity, Choice and Opportunity Cost

- Thus, when individuals decide on how to use scarce resources, they do so on the basis of *opportunity cost*.

- Every scarce good has an *opportunity cost*
  - because the production of every good or service uses resources that could have been used to produce something else
  - the opportunity cost of working overtime is the leisure that you must forgo

# The Production Possibilities Curve

- The choices that individuals/society face can be represented using ***Production Possibility Curve*** *(PPC)*
- A *PPC* describes
  - the various combinations of final goods or services that could be produced in each time period with all available resources and technology fully and efficiently employed.

- Thus, the PPC shows the various possibilities for allocating scarce resources.

# The Production Possibilities Curve

- It illustrates the concepts of **scarcity**, **choice** and **opportunity cost**.
  - **Scarcity** is implied by the unattainable combinations beyond the curve;
  - **Choice** by the need to choose among the attainable points on the curve;
  - **Opportunity cost** by the downward slope of the PPC.

# The Production Possibilities Curve

PPC

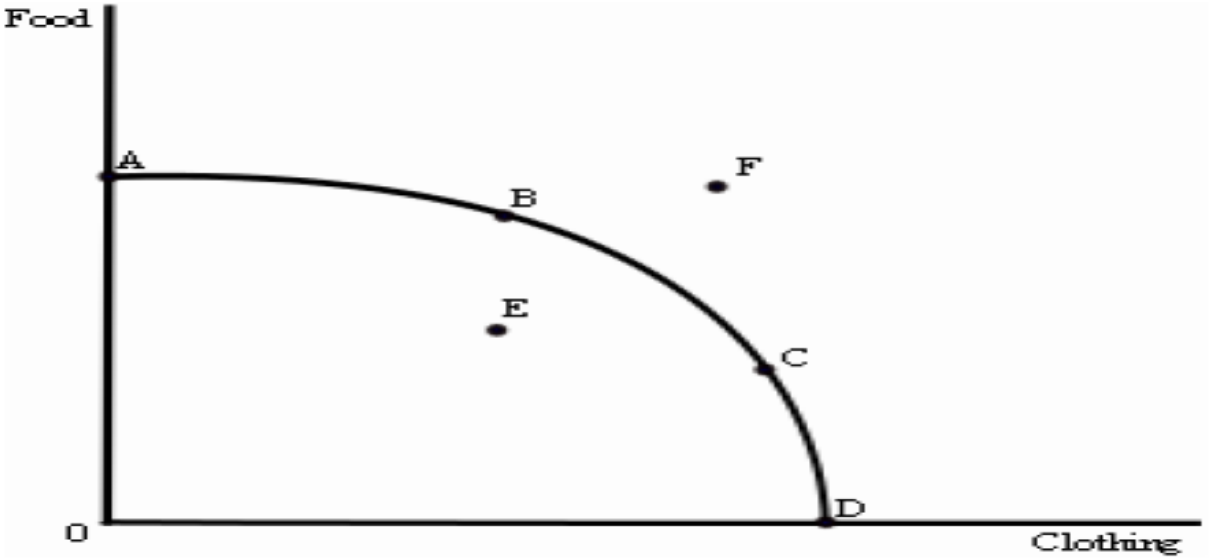



**Figure 1.1: Production Possibilities Curve**

# The Production Possibilities Curve

- The PPC shows that the economy can produce only two goods; *food* and *clothing*.
  - It shows a *trade-off* between the production of food and clothing.
- At one extreme, the country will be at point A, if it devotes all its resources to produce food.
  - This will imply it will not produce any clothing.
- On the other extreme, if it decides to devote all its resources to produce clothing it will be at point D.
  - This will imply it will not produce any food.

# The Production Possibilities Curve

- Production cannot take place beyond the PPC (e.g., point F)
  - because the country does not have enough resources to produce at F.
- It can produce at point E, but this will represent *inefficient use* and/or *underemployment* of available resources.
- Point E also implies that the economy is producing less of both goods than it is possible for it to produce.
- If the country decides to produce more food (say moving along the PPC from point D to C or from C to B)
  - it would have to ***sacrifice*** the production of some clothing.

# The Production Possibilities Curve

- This sacrifice of clothing which enables the production of more food is the *opportunity cost*.
- The PPC is *bowed outward* because the *opportunity cost* of producing more and more units of food increases as we move upwards along the curve.
- That is each additional unit of food costs the country more forgone alternatives (clothing) than the previously produced unit.

# The Production Possibilities Curve

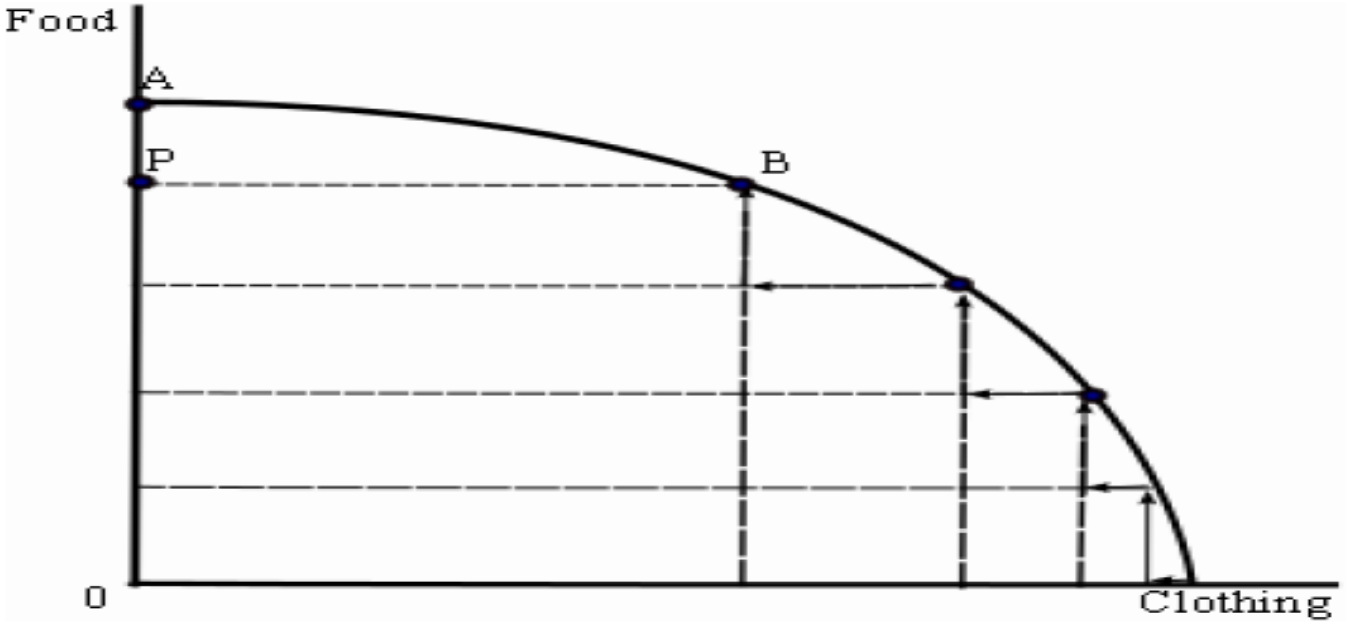



**Figure 1.2: The Law of Increasing Opportunity Cost**

# The Production Possibilities Curve

- Although each increment of food is the same, the sacrifices made (i.e. the amount of clothing forgone) in producing these increases (as seen by the arrows).
  - additional production of one good involves increasing sacrifices of the other or
  - the *opportunity cost* for each additional unit produced increases at an increasing rate.
  - This is ***the law of increasing opportunity cost***.

# The Production Possibilities Curve

- Reason for this outcome:
  - some resources are better suited for the production of some goods than others
    - For example, skills differ among people, raw materials differ from one another
- We can have more of everything
  - when resources previously unemployed are now employed, resources previously used inefficiently are now being used efficiently

# The PPC and Economic Growth

- When economic growth shifts the (PPC) outward.
  - Over time investment in new plant will increase the capital stock
  - New and more efficient raw materials may become available,
  - Technological progress is more likely to take place,
  - Through education and training labour may become more productive.
- All these factors will allow potential output to increase.

# The Production Possibilities Curve

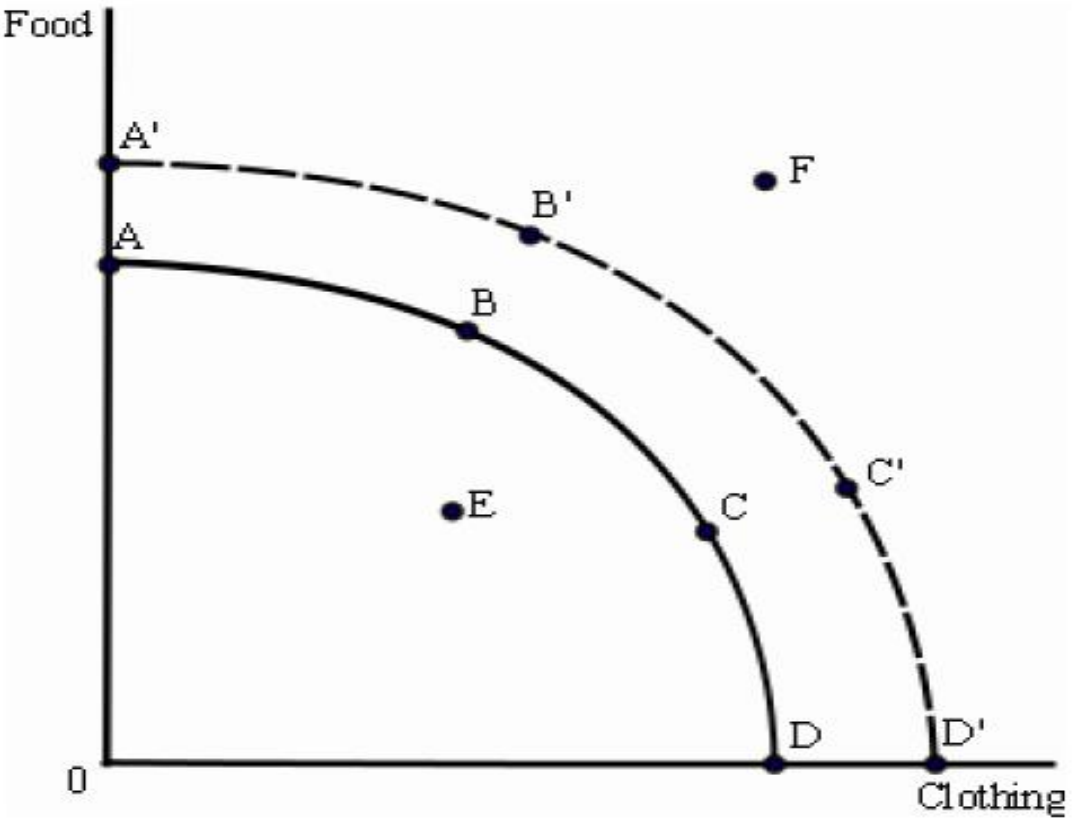



**Figure 1.3: An Outward Shift in the Production Possibility Curve**

# Microeconomics and the Economic Problem

- Microeconomics is concerned with very specific questions or decisions involving individual markets, firms, industries and consumers.
  - How do petrol taxes affect petrol sales?
  - How many students will choose not to go to university if the students' loan scheme is abolished?
  - How much will the price of yam increase if floods destroy half of the yam crop?

# Microeconomics and the Economic Problem

- Microeconomics also analyses market structures; why some markets are highly competitive.
- The study of microeconomics helps us to answer the *what*, *how* and *for whom* questions.
- But how satisfactorily these questions will be answered will depend on the society's objectives.
- These objectives are broadly grouped into
  - *economic efficiency* and *equity*

# Microeconomics and the Economic Problem

- ***Economic efficiency*** is achieved when each good is produced at the minimum cost and where individuals and firms get the maximum benefits from their resources.
  - It must involve efficiency in production (productive efficiency) and efficiency in consumption and specialization and exchange (allocative efficiency)

- ***Equity*** is where the resulting income out of production is distributed in a way that is considered to be just or fair.

# Economic Units

- Economic units are consolidated into three important groups namely; Households, Business Firms, Governments

- Households are the principal owners of factors of production.

- *Household* is broadly defined as all the people who live under one roof and who take joint financial decisions.

# Economic Units

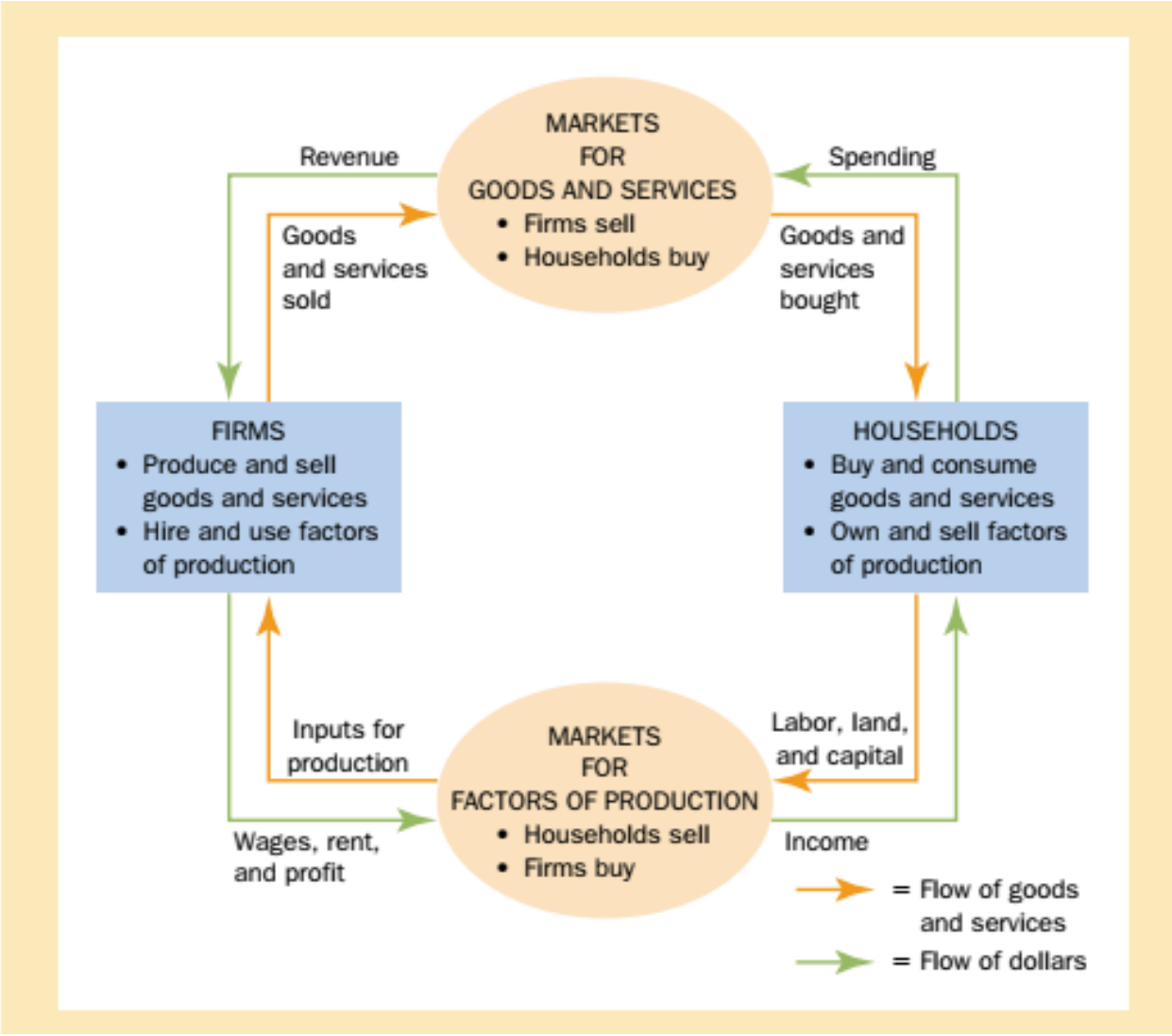
- A *firm* is defined as the unit that employs factors of production to produce commodities and/or services that it sells to other firms, to households or to the government.

- *Government* includes all the public agencies, government bodies and other organizations (institutions) belonging to the central and local government. Government obtains its revenue through taxes collected and other fee charges.

- The relationship between economic units is two-sided and can be represented in a flow diagram

# The circular flow of goods and incomes

THE CIRCULAR FLOW. This diagram is a schematic representation of the organization of the economy. Decisions are made by households and firms. Households and firms interact in the markets for goods and services (where households are buyers and firms are sellers) and in the markets for the factors of production (where firms are buyers and households are sellers). The outer set of arrows shows the flow of dollars, and the inner set of arrows shows the corresponding flow of goods and services.



# Economic Systems

- Economies are faced with the problem of *scarcity* and *choice*.
- *Government*, *firms* and *households* are also faced with how to answer the fundamental questions of resource allocation
  - what to produce
  - how to produce
  - for whom to produce
- These questions are answered by the kind of economic system in operation.
- The kinds of economic systems are, the free market economy, centrally planned economy, and mixed economy

# Economic Systems

- *Centrally planned economy*
  - Here all the economic decisions and choices about resource allocation are taken by the government.
  - Ownership and control of resources are also vested in the state in trust for the people.
- Because the free market can fail and government has to intervene to rectify the various failings of the free market gave rise to a third economic system,
  - The mixed economy - which is a hybrid of these two extreme economic systems.

# Economic Systems

- Advantages of a command economy
  - high investment, high growth
  - stable growth
  - social goals pursued
  - low unemployment
- Problems of a command economy
  - problems of gathering information
  - expensive to administer
  - inappropriate incentives
  - shortages and surpluses

# Economic Systems

- *Free market economy* (a *laissez faire* economy)
  - Here, government normally does not play an active role in the economy.
  - Decisions and choice about resource allocation are left to market forces of supply and demand
  - The demand and supply decisions of consumers and firms are transmitted to each other through their effect on prices: **through the price mechanism**.
  - Prices respond to shortages and surpluses. Shortages result in prices rising. Surpluses result in prices falling.
  - The price at which demand equals supply is called **equilibrium price.**

# The price mechanism: the effect of a rise in demand

## Goods Market

$D_g$ → shortage ($D_g > S_g$) → $P_g$ ↑ → $S_g$ ↑ / $D_g$ ↓ until $D_g = S_g$

## Factor Market

$S_g$ → $D_f$ → shortage ($D_f > S_f$) → $P_f$ ↑ → $S_f$ ↑ / $D_f$ ↓ until $D_f = S_f$

# Economic Systems

- *Free market economy* (a *laissez faire* economy)
  - Here, government normally does not play an active role in the economy.
  - Decisions and choice about resource allocation are left to market forces of supply and demand
  - Government's main role is to facilitate the operations of the system by
    - maintaining an *effective security* and *legal system*
    - Providing the basic social infrastructure and good economic environment for the smooth running of businesses.

# Economic Systems

- *Mixed economy*
- Here, decisions and choices are made partly by free market forces of demand and supply and partly by government.
- It is important to note that all economies are mixed, though with different degree of free market and centrally planned decision-making.

# THE NATURE OF ECONOMIC REASONING

- Economics as a science (predict and explain)
  - models in economics ( part or whole economy)
  - building models (hypotheses to explain causes of economic situation-induction)
  - Deduction: drawing conclusion from a model
  - using models (explain and predict)
  - assessing models (accurate predictions)

# THE NATURE OF ECONOMIC REASONING

- Economics as a social science
  - –difficulties in conducting controlled experiments in many parts of the subject
  - –problems of predicting human behaviour

# Economics and policy

- Economists play a major role in helping governments to devise economic policy.
- To understand this role, it is necessary to distinguish between positive and normative statements.
- A positive statement is a statement of fact. It may be right or wrong, but its accuracy can be tested by appealing to the facts.
- A normative statement is a statement of value: a statement about what ought (should) or ought not to be, about whether something is good or bad, desirable or undesirable.
- Economics can only contribute to policy issues in a positive way. Economists, as scientists, should not make normative judgements. They can make them only as individual people, with no more moral right than any other individual

# The End



## Next Time: Demand and Supply Analysis